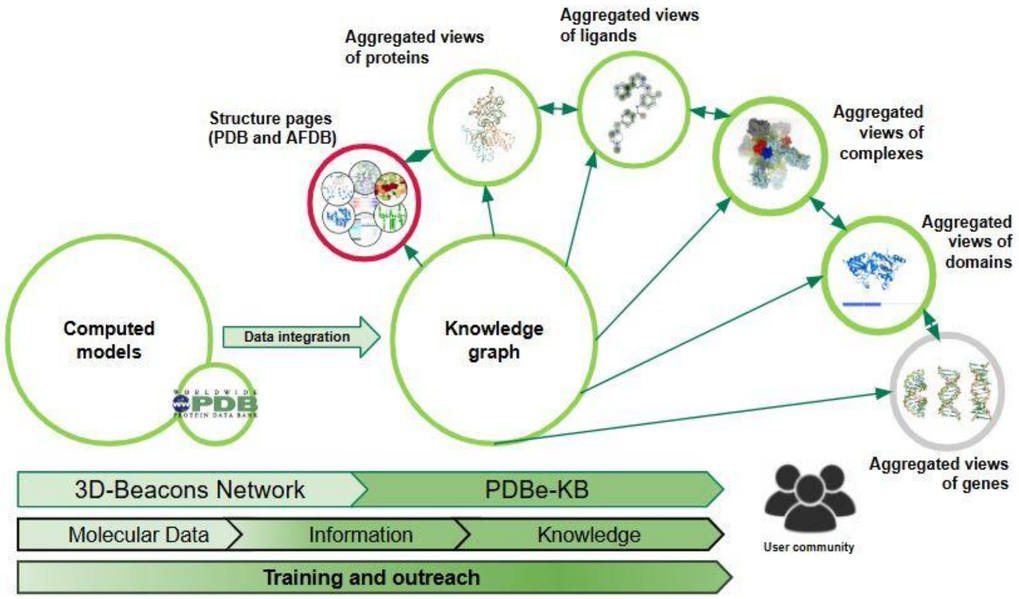
Enhancing Structural Data Connectivity in the Protein Data Bank

Sudakshina Ganguly[1], Jennifer Fleming[1], Ivanna Pidruchna[1], Dare Kayode Lawal[1], Marcus Bage[1], Adam Midlik[1], Preeti Choudhary[1], Sreenath Nair[1], Marcelo Querino Lima Afonso[1], Ibrahim Roshan Kunnakkattu[1], Sri Devan Appasamy[1], Sameer Velankar[1]

(1) EMBL EBI

PDBe Protein Data Bank in Europe

EMBL-EBI



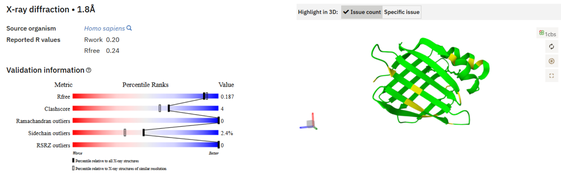
- This project will develop an **integrated web portal** providing access to structures, enriched annotations & aggregation of structure data to provide additional information.
- The >200K entries represent ~61K unique proteins (on average >3 structures for each protein)
- Our users come from all existing generations with very different backgrounds, expertises and data needs: Bioinformaticians, Computer Scientists, Curators, Structural Biologists, Medicinal Chemists
- Our mission to establish a high-quality, up-to-date, and integrated infrastructure, along with innovative data analysis tools for 3D macromolecular structures.

New changes to the current pages are being implemented

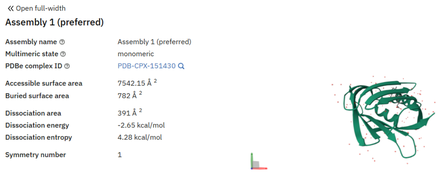
- Top horizontal tabs for in-page navigation.
- 3D viewer on the right to give the user overview of the structure

Overall information about the entry under the **Summary** tab

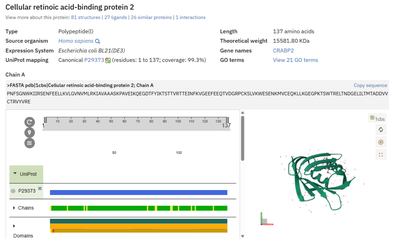
Information about the validation metrics including 3D view with labelling according to model quality issues under the **Model Quility** tab



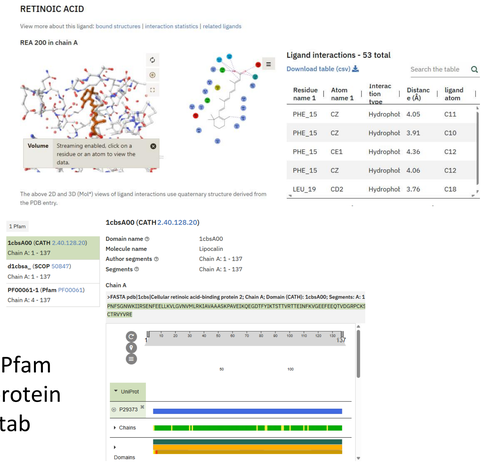
3D view of the preferred assemblies within the entry under the **Assemblies** tab



Overall details about each macromolecule within the entry under the **Macromolecules** tab



Information about ligands and the interaction view within the binding site under the **Ligands and Environments** tab



Annotations from SCOP, CATH, Pfam and other databases regarding protein domains under the **Domains** tab

EMBL's European Bioinformatics Institute (EMBL-EBI)
Wellcome Genome Campus, Hinxton,
Cambridgeshire. CB10 1SD. UK.
T +44(0)1223 494 444